

daa lab 01.py

daa lab 02.py

daa lab 04.py

daa lab 03.py

Run and Debug

To customize Run and Debug, open a folder and create a launch.json file.

Show automatic Python configurations

42 def binary_search(arr, target):

43 """

44 Binary Search Algorithm

45 """

46 low, high = 0, len(arr) - 1

PROBLEMS 12

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 12

Python Debug Console

abınayasrınıvasan@abınayas-MacBook-Aır ~ % /usr/bin/env /usr/local/bin/python3.14 /Users/abınayasrınıvasan/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51239 -- /Users/abınayasrınıvasan/daa\ lab\ 01.py

Array: [2, 5, 10, 15, 23, 35, 48, 60, 75, 90, 105, 120]

Searching for: 35

Found at index: 5, Comparisons: 3

Size	IS Time(ms)	BS Time(ms)	IS Comparisons	BS Comparisons
1000	0.0006	0.0019	1	8
5000	0.0006	0.0011	4	11
10000	0.0005	0.0009	4	13
50000	0.0005	0.0008	3	16
100000	0.0006	0.0009	4	17

abınayasrınıvasan@abınayas-MacBook-Aır ~ %

BREAKPOINTS

☐ Raised Exceptions

☒ Uncaught Exceptions

☐ User Uncaught Exceptions

PYTHON RESOURCE MONITOR

Launchpad

0 0 12

Java: Ready

Sourcery Analytics

Ln 108, Col 23

Spaces: 4

UTF-8

LF

{ } Python

Python 3.14.3



- 


```

1 // 1. Sum of first 10 natural numbers
2 #include <iostream>
3 using namespace std;
4
5 int main()
6 {
7     int sum = 0;
8     for (int i = 1; i <= 10; i++)
9     {
10         sum += i;
11     }
12     cout << "Sum of first 10 natural numbers is: " << sum << endl;
13     return 0;
14 }

```

 Python Debug Console ... |

Pattern	Naive	KMP	RK
AB	12536	10000	1160
ABCD	13268	10000	211
ABCDAB	13301	10005	117
ABCDABCD	13301	10006	134

Ln 145, Col 49 Spaces: 4 UTF-8 LF {} Python Python 3.14.3

Run and Debug

To customize Run and Debug, **open a folder** and create a launch.json file.

Show automatic Python configurations

Users > abinaysrinivasan >  daa lab 03.py >  UnionFind >  union

PROBLEMS 12

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 12

 Python Debug Console ... |

```

● abinayasrinivasan@abinayas-MacBook-Air ~ % /usr/bin/env /usr/local/bin/python3.14 /Users/abinayasrinivasan/.vscode/extensions/ms-python.debugpy-2026.6.0-d
arwin-arm64/bundled/libs/debugpy/adapters/..../debugpy/launcher 51316 -- /Users/abinayasrinivasan/daa\ lab\ 03.py

```

```

===== Kruskal's MST =====
Edge (0 - 3) Weight: 5
Edge (2 - 4) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 6) Weight: 9
Total MST Cost: 39

```

```

=== Prim's MST ===
Edge (0 - 3) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 2) Weight: 5
Edge (4 - 6) Weight: 9
Total MST Cost: 39

```

```

Total NSR cost: 35
○ abinayasrinivasan@abinayas-MacBook-Air ~ %

```

▼ BREAKPOINTS

- ☐ Raised Exceptions
- ☒ Uncaught Exceptions
- ☐ User Uncaught Exceptions

> PYTHON RESOURCE MONITOR

daa lab 04.py

daa lab 01.py daa lab 02.py daa lab 04.py x daa lab 03.py

Run and Debug

To customize Run and Debug, open a folder and create a launch.json file.

Show automatic Python configurations

Users > abinayasrinivasan > daa lab 04.py > ...

1 import heapq

2

3

4 def dijkstra(graph, source):

5

PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 12

Python Debug Console

abinayasrinivasan@abinayas-MacBook-Air ~ % /usr/bin/env /usr/local/bin/python3.14 /Users/abinayasrinivasan/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51338 -- /Users/abinayasrinivasan/daa\ lab\ 04.py

Shortest paths from vertex 0:

Vertex	Distance	Path
0	0	0
1	3	0 -> 2 -> 1
2	1	0 -> 2
3	4	0 -> 2 -> 1 -> 3
4	7	0 -> 2 -> 1 -> 3 -> 4
5	9	0 -> 2 -> 1 -> 3 -> 4 -> 5

abinayasrinivasan@abinayas-MacBook-Air ~ %

BREAKPOINTS

☐ Raised Exceptions

☒ Uncaught Exceptions

☐ User Uncaught Exceptions

PYTHON RESOURCE MONITOR

Launchpad 0 0 12 Java: Ready Sourcery Analytics

Ln 2, Col 1 Spaces: 4 UTF-8 LF Python Python 3.14.3